

Design FMEA — VBF-T6R2-DFMEA-001

Design FMEA — LNMO/Graphite 4C Smartphone Cell (VBF-T6R2-DFMEA-001)

****Case**:** exp/t6_r2 · ****Candidate**:** R12a_h140 · ****Date**:** 2026-08-26
Severity (S) / Occurrence (O) / Detection (D): 1–10 scale; RPN = S·O·D. Mitigations reflect the design decisions actually simulated (log rounds R1–R12).

#	Failure mode	Effect
1	Lithium plating at 4C charge	Capacity loss, dendrite risk, safety
2	Overheating during 4C charge in hot ambient	T > 50 °C, thermal aging
3	Thermal-interface degradation	T margin loss → criteria breach
4	SEI overgrowth > 500 nm by 100 cycles	Impedance growth, capacity fade
5	Effective C-rate mismatch	Verification gap: 4C judged at 1C
6	Cathode particle cracking / fading (long term)	Capacity fade beyond 100 cycles
7	Electrolyte oxidation at 4.7 V	Gas generation, impedance growth
8	LNMO file parameterization artifact	Wrong diffusion input, misleading

Top RPN: #6 (long-term cracking, not simulatable in this parameter set) and #2/#7 (thermal margin and electrolyte oxidation). Recommended follow-ups: true 25.5 A DFN validation, material-stage cracking screening, and real_compute=true endorsement for paper-grade claims.